

Jun Seok Kim

ijunsuk9797@gmail.com
forestinblue.github.io • github.com/forestinblue

RESEARCH INTERESTS

- 3D scene representation for robotics
- 3D perception under compute and communication constraints
- Real-time reconstruction of dynamic scenes
- Uncertainty calibration for learned perception

EDUCATION

The Chinese University of Hong Kong, Shenzhen

Master of Science in Artificial Intelligence and Robotics

Capstone Project : LocalAggTR (Grade: Distinction)

Sep 2024 – Expected Mar 2027

Kangwon National University

Bachelor of Science, Information Statistics

Bachelor of Science in Engineering, Convergence Program of Data Science (Double Major)

GPA: 3.63/4.5 (90.1%)

Main Courses: Machine Learning · Theory of Probability · Multivariate Statistical Analysis

Mar 2017 – Feb 2023

PUBLICATIONS & PREPRINTS

[1] **J. Kim** and R. Huang. Range-aware uncertainty calibration in self-supervised monocular depth.

In preparation for CVPR 2027

RESEARCH EXPERIENCE

MSc Student Researcher, Computer Vision Laboratory (Advisor: Prof. [Rui Huang](#))

Sep 2024 – Present

Range-aware uncertainty calibration in self-supervised monocular depth (MSc thesis) Jan 2026 – Present

- **Robots act on the uncertainty** of self-supervised depth models, yet it is evaluated only by ranking, so its magnitude goes unchecked. On KITTI, MonoProb (WACV 2024) underestimates its error at every range, with error reaching $1.9\times$ the predicted uncertainty.
- Showed that self-supervised retraining alone cannot close this gap, while a single **label-free** scale constant estimated from stereo residuals can. Retrained only the uncertainty head with a second self-supervised model as the teacher, then applied that constant.
- Cut worst-range **calibration** error by 89% versus the original model and by 74% versus rescaling fitted on LiDAR ground truth, using **no ground-truth depth**.

LocalAggTR: Differentiable Voxelization for Weakly Supervised 3D Occupancy

Sep 2025 – Dec 2025

[\[code\]](#) · Capstone project

- GaussTR's Gaussian-to-voxel step is non-differentiable and dominates inference time, so 3D occupancy losses cannot train the network end-to-end.
- Replaced it with a differentiable **Local Aggregation** operator that passes occupancy gradients.
- Made **inference $7.2\times$ faster** with mIoU unchanged on nuScenes. **Adopted by the lab's 3D Gaussian Splatting project** for semi-supervised fine-tuning.

Point Cloud Density and 3D Object Detection

Jun 2025 – Jul 2025

[\[code\]](#)

- Asked how a **3D object detector** degrades when LiDAR density drops and whether upsampling recovers it.
- Built a resampling pipeline that thins LiDAR points by random or voxel-grid downsampling and restores them with two upsampling methods, then evaluated PointPillars on nuScenes-mini.
- Found that how points are removed matters more than how many. Voxel-grid thinning to one fifth of the points cost 2.1% of mAP, while random thinning to half cost about 4× more.

Geometric vs. Generative 3D Augmentation for Point Cloud Classification

Jan 2025 – May 2025

[\[code\]](#)

- Tested whether text-to-3D generative augmentation outperforms geometric augmentation.
- Built Point-E generation and geometric augmentation modules and ablated them with PointNet on ModelNet40.
- **Geometric augmentation** raised accuracy by 6.5 points, while text-to-3D generative augmentation did not help.

INDUSTRY EXPERIENCE

Founder, [TARS](#)

Feb 2024 – Present, South Korea

- Built and operate a **YOLO-based foot-traffic analytics** service for small businesses such as restaurants, cafés and real-estate offices, measuring hourly pass-by, stop and entry rates from storefront cameras.
- Serve 60+ paying clients with consulting reports based on this data. Self-funded and profitable since launch.

Backend & Infrastructure Engineer, [Bluzen Corp.](#)

Jul 2022 – Aug 2022, Singapore

- Built a MariaDB and Node.js backend for workforce records and networked Hyundai Motor server rooms.

Data Scientist, [Connple Corp.](#)

Jun 2021 – Jun 2022, South Korea

- Built and shipped a **BERT-based restaurant recommender** to a **production mobile app**, paired with SVD collaborative filtering on review ratings.
- Extracted food keywords from Instagram reviews with LDA topic modeling to power a food discovery service.
- Automated nationwide collection of 600,000 restaurant records through a scheduled Flask data pipeline.

SCHOLARSHIPS AND HONORS

- **Mid-Sized Company Employment Support Scholarship**, Korea Student Aid Foundation (KOSAF) 2021–2023
- **Passion Award**, 21st & 22nd Gangwon Province University Student Entrepreneurship Competition 2020, 2021
- **Commendations for exemplary service and field construction**, Republic of Korea Army 2019

LEADERSHIP

- **President, Photo Club**, Kangwon National University, 100+ members Jan 2021 – Jun 2021
- **Team Lead, Student Startup Teams**, Kangwon National University Jan 2020 – Dec 2021
- **Squad Leader, 930th Artillery Battalion**, Republic of Korea Army Apr 2018 – Nov 2019

TEACHING EXPERIENCE

- **Teaching Assistant**, Introduction to Python, Kangwon National University Mar 2020 – Jun 2020

SKILLS & LANGUAGES

- **Programming:** Python, C/C++, CUDA, SQL
- **Frameworks:** PyTorch, MMDetection3D, MMEEngine, scikit-learn
- **Tools:** Git, Linux, Docker
- **Languages:** Korean (native), English (full professional), Mandarin Chinese (basic)